

QUADRATIC-STATE SHEAR OVER $\mathbb{F}_2$: EXACT FIBRES AND THE COMPLETE WITT-SENSITIVE FUNCTIONAL GRAPH

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ABSTRACT. Let (V, Q) be a nonsingular quadratic space of dimension $2m$ over $\mathbb{F}_2$, and iterate

$$\Phi(x, y) = (y, x + Q(x)y)$$

on V^2 . We determine its complete functional graph for both Witt signs. The polar bit is invariant, and a three-bit quotient shows that every transient has depth at most two and every eventual period belongs to $\{1, 2, 3, 4\}$, with all pointwise exceptions classified. A two-candidate inverse formula gives every fibre, the full image tower, and the exact three depth layers. Character sums then resolve the Witt dependence, while the reverse fibres force precisely six decorated cycle shapes. This yields closed counts for every component and cycle, hence the finite-map zeta function. The map is nonbijective and fails both standard Yang–Baxter relations; transvection actions, Yang–Baxter pair maps, static quadratic-form counts, and zeta bookkeeping are therefore treated only as owner boundaries. Exact exhaustive controls through dimension ten include literal image-set equality and component traversal. The owner search is bounded, and no novelty, priority, or external-release claim is made.

1. THE MAP AND THE RESIDUAL QUESTION

Let (V, Q) be a nonsingular quadratic space of dimension $2m$ over $\mathbb{F}_2$, with polar form

$$B(x, y) = Q(x + y) + Q(x) + Q(y).$$

On V^2 , define

$$(1.1) \quad \Phi(x, y) = (y, x + Q(x)y).$$

The update is basis-free and equivariant for the diagonal action of $O(V, Q)$. It swaps the pair when the departing vector is singular and adds that vector to its partner when it is nonsingular.

Put

$$(1.2) \quad N = |V| = 2^{2m}, \quad S = \sum_{x \in V} (-1)^{Q(x)} = \varepsilon 2^m, \quad \varepsilon \in \{+1, -1\}.$$

For $m = 0$, only the plus convention occurs and the unique state is fixed. For $m \geq 1$, the two signs are the two Witt types. Classical characteristic-two quadratic counts and orthogonal geometry are background [4, 5]. We prove the counts needed below, so no static census receives contribution credit.

The word “shear” does not identify (1.1) with a transvection. A symplectic transvection has a fixed centre and coefficient supplied by the polar pairing; its generated group acts bijectively [6]. Here the moving state supplies $Q(x)$, the active vector changes at every step, and the map is noninjective. Set-theoretic Yang–Baxter maps are another pair-map neighbor, including both bijective [3] and nonbijective finite-order settings [2]. We give explicit failures of both Yang–Baxter relations in remark 2.3.

After these deductions, the residual package is the conjunction of exact pointwise dynamics, every fibre and image, Witt-sensitive temporal layers, and all decorated components of the literal map. The finite-map zeta is a consequence, not a separate contribution.

There are also internal package collisions. The following firewall records them without claiming an external literature comparison.

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Neighbor	Subtraction and distinction
P99	A fixed unipotent matrix acts bijectively on integer sublattices; here the shear is state-gated and nonbijective.
P103	Double-adjugate dynamics uses rank strata and scalar power maps; here the engine is the quadratic/polar quotient and nonuniform fibres.
P106	Its cubic polarity identity gives depth at most one and periods at most two; this is the closest temporal silhouette, but (1.1) has a Witt-sensitive depth-two layer and periods three and four.
P109	Nilpotent image acts on a subspace lattice with one recurrent state; (1.1) acts on vector pairs and has four cycle lengths.
P118	Quotient, fibre, basin, and zeta packaging is already spent; its carrier and labelled-EGF fibre engine are unrelated.

2. POINTWISE TEMPORAL CLASSIFICATION

For a state (x, y) , set

$$a = Q(x), \quad b = Q(y), \quad c = B(x, y).$$

Proposition 2.1 (three-bit quotient). *The polar bit c is invariant, and*

$$(2.1) \quad (a, b, c) \mapsto (b, a(1 + b + c), c).$$

Proof. Writing $(u, v) = \Phi(x, y)$, polarization and alternation give

$$Q(u) = b, \quad Q(v) = Q(x + ay) = a + ab + ac = a(1 + b + c), \quad B(u, v) = B(y, x + ay) = c.$$

□

Theorem 2.2 (complete pointwise clock). *Every transient has depth at most two, and every eventual period belongs to $\{1, 2, 3, 4\}$. More precisely:*

c	(a, b)	pointwise fate
0	00	recurrent; period 1 iff $x = y$, otherwise 2
0	01	recurrent; period 2 iff $x = 0$, otherwise 4
0	10	recurrent; period 2 iff $y = 0$, otherwise 4
0	11	depth 1; eventual period 2 iff $x = y$, otherwise 4
1	00	recurrent of exact period 2
1	01	depth 2, followed by exact period 2
1	10	depth 1, followed by exact period 2
1	11	recurrent of exact period 3

Depth two occurs in the plus plane for $m = 1$, in both signs for $m \geq 2$, and not in the minus plane for $m = 1$.

Proof. Equation (2.1) gives, for $c = 0$,

$$00 \rightarrow 00, \quad 01 \leftrightarrow 10, \quad 11 \rightarrow 10,$$

and, for $c = 1$,

$$00 \rightarrow 00, \quad 01 \rightarrow 10 \rightarrow 00, \quad 10 \rightarrow 00, \quad 11 \rightarrow 11.$$

On the pair of vectors, a singular or nonsingular departing state acts by

$$A_0 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad A_1 = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \quad \text{over } \mathbb{F}_2.$$

In particular,

$$A_0^2 = I, \quad A_1^3 = I, \quad A_1 A_0 = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad A_0 A_1 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},$$

and both two-step products square to I . The matrix word or landing test for every quotient row is therefore as follows.

c	start	word and pointwise consequence
0	00	$A_0^2 = I$; one step fixes the pair exactly when $x = y$.
0	01	$(A_1 A_0)^2 = I$; two steps fix the pair exactly when $x = 0$.
0	10	$(A_0 A_1)^2 = I$; two steps fix the pair exactly when $y = 0$.
0	11	A_1 lands in type 10 with second coordinate $x + y$; hence the eventual period is two exactly when $x = y$.
1	00	$A_0^2 = I$, while $\text{Fix}(A_0)$ would force $x = y$, excluded by $B(x, y) = 1$.
1	01	$A_1 A_0$ lands in type 00 after two steps, whose period is exactly two.
1	10	A_1 lands in type 00 after one step, whose period is exactly two.
1	11	$A_1^3 = I$, and $\text{Fix}(A_1) = \{(0, 0)\}$, excluded by this type.

These products give all listed periods and the only possible shortenings $x = y$, $x = 0$, or $y = 0$. Thus no further period or transient occurs.

The depth-two type has the cardinality computed as A in (3.3); the stated small-form boundary follows immediately. $\square$

Remark 2.3 (not a Yang–Baxter map). In a plus hyperbolic plane choose e, f with $Q(e) = Q(f) = 0$, $B(e, f) = 1$. Directly,

$$\Phi_{12}\Phi_{23}\Phi_{12}(e + f, e, f) = (f, e, f), \quad \Phi_{23}\Phi_{12}\Phi_{23}(e + f, e, f) = (f, e, e).$$

Thus the braid relation fails. With products acting right-to-left, the two sides of $R_{12}R_{13}R_{23} = R_{23}R_{13}R_{12}$, evaluated at $(e + f, e + f, e)$, are (e, f, e) and $(e, f, 0)$. Hence the quantum Yang–Baxter convention fails as well.

3. FIBRES, IMAGES, AND WITT LAYERS

Theorem 3.1 (every fibre). *For each target $(u, v) \in V^2$,*

$$(3.1) \quad \Phi^{-1}(u, v) = \{(v, u) : Q(v) = 0\} \cup \{(u + v, u) : Q(u + v) = 1\}.$$

The two candidates are distinct whenever both occur. Consequently the numbers of targets of indegree 0, 1, 2 are

$$(3.2) \quad d_0 = d_2 = \frac{N(N-1)}{4}, \quad d_1 = \frac{N(N+1)}{2}.$$

Proof. An unknown preimage has the form (x, u) , with $v = x + Q(x)u$. The alternatives $Q(x) = 0, 1$ give exactly (3.1), and substitution proves sufficiency. Equality of the two candidates would force $u = 0$, contradicting their simultaneous conditions $Q(v) = 0$ and $Q(v) = 1$.

The change of variables $(u, v) \leftrightarrow (v, w = u + v)$ is a bijection. A missing target has $Q(v) = 1, Q(w) = 0$, while a double target has $Q(v) = 0, Q(w) = 1$. Since the numbers of singular and nonsingular vectors are $N_0 = (N + S)/2$ and $N_1 = (N - S)/2$, either class has size $N_0 N_1 = (N^2 - S^2)/4 = N(N - 1)/4$. Edge mass gives d_1 . $\square$

To enumerate the temporal types, let

$$C_{abc} = \#\{(x, y) : Q(x) = a, Q(y) = b, B(x, y) = c\}.$$

Lemma 3.2 (pair census). *The eight counts reduce to*

$$(3.3) \quad \begin{aligned} H = C_{000} &= \frac{N(N + 3S + 4)}{8}, \\ M = C_{010} = C_{100} = C_{110} &= \frac{N(N - S)}{8}, \\ A = C_{001} = C_{011} = C_{101} &= \frac{N(N + S - 2)}{8}, \\ Z = C_{111} &= \frac{N(N - 3S + 2)}{8}. \end{aligned}$$

Proof. Expanding the three binary indicators gives

$$(3.4) \quad \begin{aligned} C_{abc} &= \frac{1}{8} [N^2 + (-1)^a NS + (-1)^b NS + (-1)^{a+b} S^2 \\ &\quad + (-1)^c N \{1 + (-1)^a + (-1)^b + (-1)^{a+b} S\}]. \end{aligned}$$

Indeed, nondegeneracy gives the three mixed polar sums N, N, N , while $Q(x) + Q(y) + B(x, y) = Q(x + y)$ gives the final sum NS . Now use $S^2 = N$. This self-contained Walsh calculation is retained even though such static quadratic counts belong to the classical owner region. $\square$

Theorem 3.3 (image tower and exact layers). *The second image is exactly the recurrent set, and*

$$(3.5) \quad |\operatorname{im} \Phi^0| = N^2, \quad |\operatorname{im} \Phi| = \frac{N(3N+1)}{4}, \quad |\operatorname{im} \Phi^t| = \frac{N(5N-S+4)}{8} \quad (t \geq 2).$$

The numbers L_j of states of depth j are

$$(3.6) \quad L_0 = \frac{N(5N-S+4)}{8}, \quad L_1 = \frac{N(N-1)}{4}, \quad L_2 = \frac{N(N+S-2)}{8}.$$

Proof. Formula (3.1) has values, on the four (a, b) -columns,

$$\begin{array}{c|cccc} c & 00 & 01 & 10 & 11 \\ \hline 0 & 1 & 1 & 2 & 0 \\ 1 & 2 & 0 & 1 & 1 \end{array}.$$

Together with theorem 2.2, this says that the first image is the recurrent set plus type $(c, a, b) = (1, 1, 0)$. That extra type maps to the recurrent type $(1, 0, 0)$, while recurrent points have predecessors of every iterate along their cycles. Hence $\operatorname{im} \Phi^2 = \operatorname{Rec}(\Phi)$ setwise. The first image size follows from (3.2); reading the transient rows and substituting (3.3) gives $L_2 = A$, $L_1 = M + A$, and the displayed formulas. $\square$

4. ALL COMPONENTS AND THE ZETA FUNCTION

The reverse dynamics is complementary to the forward quotient: it determines the trees attached to the cycles already classified in theorem 2.2.

Theorem 4.1 (six decorated component shapes). *Every connected component has exactly one of the following shapes.*

Shape	Number
bare fixed point	N_0
bare 2-cycle	$(H - N_0)/2$
2-cycle with one leaf at one cycle vertex	N_1
2-cycle with one length-two tail at each cycle vertex	$A/2$
bare 3-cycle	$Z/3$
4-cycle with one leaf at each of two alternating vertices	$(M - N_1)/2$

There are no other reverse trees.

Proof. In $c = 0$, type 00 is the swap: its N_0 diagonal states are fixed, and the others form bare 2-cycles. Types 01 and 10 form 2-cycles when one coordinate is zero and 4-cycles otherwise. The fibre table in the proof of theorem 3.3 gives every type-10 cycle vertex one additional type-11 leaf. Thus the exceptional 2-cycles number N_1 ; the remaining mixed states give $(M - N_1)/2$ four-cycles, with leaves at alternating vertices.

In $c = 1$, each type-00 state lies on a 2-cycle and has one extra type-10 predecessor; that predecessor has one type-01 predecessor. This gives $A/2$ components with a length-two tail at both cycle vertices. Type 11 gives $Z/3$ bare 3-cycles. The zero fibres of the leaf types and the global indegree ceiling two exclude every deeper or branched tree. $\square$

Corollary 4.2 (cycles and zeta). *If c_j is the number of cycles of exact length j , then*

$$(4.1) \quad \begin{aligned} c_1 &= \frac{N+S}{2}, \\ c_2 &= \frac{N^2 + 2NS + 3N - 6S}{8}, \\ c_3 &= \frac{N(N-3S+2)}{24}, \\ c_4 &= \frac{N^2 - NS - 4N + 4S}{16}. \end{aligned}$$

The finite-map zeta function is

$$(4.2) \quad \zeta_\Phi(t) = (1-t)^{-c_1} (1-t^2)^{-c_2} (1-t^3)^{-c_3} (1-t^4)^{-c_4}.$$

Proof. Sum the corresponding rows of theorem 4.1 and substitute (3.3). The standard periodic-point definition and cycle product of Artin–Mazur zeta functions [1] give (4.2). $\square$

5. EXACT CONTROLS AND RELEASE BOUNDARY

The paper-local standard-library verifier exhausts the zero-dimensional boundary and both Witt signs for $1 \leq m \leq 5$, hence every ordered pair through dimension ten. It checks the literal update, polarization, every quotient transition, every pointwise depth and period, all target fibres, all eight pair counts, the fibre histogram, image and layer formulas, and all cycle counts. It also constructs the second-image set and compares it pointwise with $\text{Rec}(\Phi)$, then traverses every functional component, canonically records the reverse rooted tree at each cycle vertex modulo directed cyclic rotation only, and checks the six shapes and their counts. An asymmetric sentinel prevents reflection identification. The canonical run makes 27,405,887 exact assertions.

Finite exhaustion proves none of the all-size arguments or owner statements. The source search included the literal formula, state-dependent shear and switch vocabulary, orthogonal and symplectic transvections, and bijective and nonbijective Yang–Baxter pair maps. No direct or conjugate-map owner was located in this bounded pass. Because (1.1) is extremely short, owner risk remains substantial: a direct prior classification would supersede the residual claim. There is no assertion of first occurrence, novelty, or priority. Public posting, submission, and external circulation remain on hold.

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